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Summarize the 1906 San Francisco Earthquake. What factors shaped the current landscape?

The **1906 San Francisco Earthquake** was one of the most devastating natural disasters in U.S. history, striking Northern California on April 18, 1906.

What happened

- A massive rupture along the **San Andreas Fault** triggered an earthquake estimated around magnitude 7.8.
- The shaking lasted less than a minute, but it caused widespread collapse of buildings across **San Francisco** and surrounding areas.
- The quake ruptured the ground for nearly 300 miles.

Why it was so destructive

Several factors made the disaster far worse than the quake itself:

- **Fires**: Broken gas lines ignited fires that burned for days, destroying much of the city. In fact, fire damage exceeded the damage from shaking.
- **Building practices**: Many structures were poorly built or made of unreinforced masonry, making them highly vulnerable.
- **Infrastructure failure**: Water mains broke, leaving firefighters unable to control the blazes.
- **Population density**: A growing urban population meant more people and property were exposed.

Human impact

- About 3,000 people were killed.
- Over 200,000 were left homeless (more than half the city's population at the time).
- Large refugee camps were set up in parks and open spaces.

How it shaped today's landscape

1. Earthquake science and awareness

- The disaster led to major advances in **seismology**.
- Scientists closely studied the **elastic rebound theory**, explaining how stress builds and releases along faults.

2. Building codes and engineering

- Modern **earthquake-resistant design** grew out of lessons learned in 1906.
- California now has some of the strictest seismic building codes in the world.

3. Urban planning changes

- The city was rebuilt with wider streets and improved infrastructure.
- Fire safety systems (hydrants, cisterns) were redesigned to prevent similar catastrophes.

4. Landscape and geography

- The quake physically reshaped parts of Northern California—offset fences, roads, and even streams along the fault line.
- Development patterns shifted, with greater awareness of fault zones and unstable ground.

5. Emergency preparedness culture

- The event created a lasting culture of disaster preparedness in California, influencing policies, drills, and public education.

In short: the 1906 earthquake wasn't just a destructive event—it fundamentally transformed science, engineering, and city planning. The modern look, safety standards, and risk awareness of San Francisco today are direct results of the lessons learned from that disaster.

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